School of Computer Science Engineering and Technology

Course-B. Tech	**Type-** Core
Course Code- CSET207	**Course Name-** COMPUTER NETWORKS
Year- 2025-26	**Semester-** Even
Date- 12-16/01/2026	**Batch-** 2024-2028

CO-Mapping

	CO1	CO2	CO3
Task 1	√		
Task 2	√		
Task 3	√		
Task 4	√		
Task 5	√	√	

Objectives

In this lab, you will learn how multiple network devices can be used to create a network in Cisco Packet Tracer (CPT). We will also learn how to configure different end devices, including the network devices for exchange of information in the network. We will also connect different end devices using various network topologies. The experiments are required to be executed with IP addresses from **A, B and C Classes** explained in Lab 1.

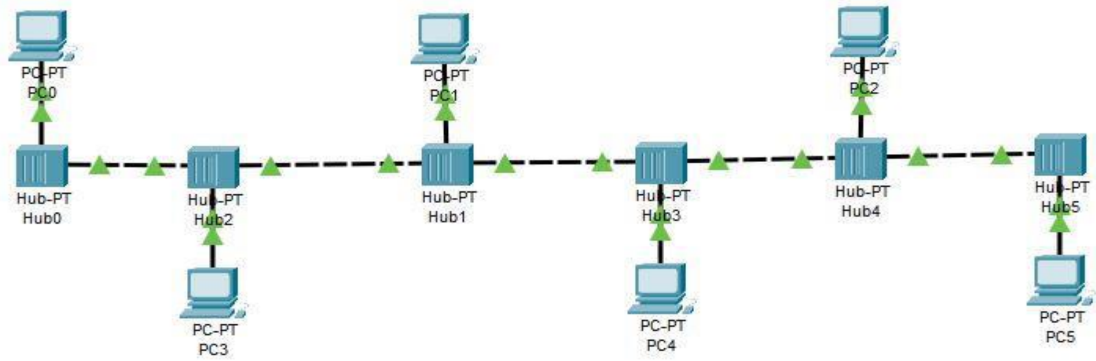
Aim

1. Understand how network Topologies work.
2. Understanding how hardware limitations and performance gains restrict us to using STAR topology and rest of the configurations are experimentation limited.
3. Understanding how IP Addresses of same network within a local point to point communication setup can communicate without and routing or static mapping assistance.

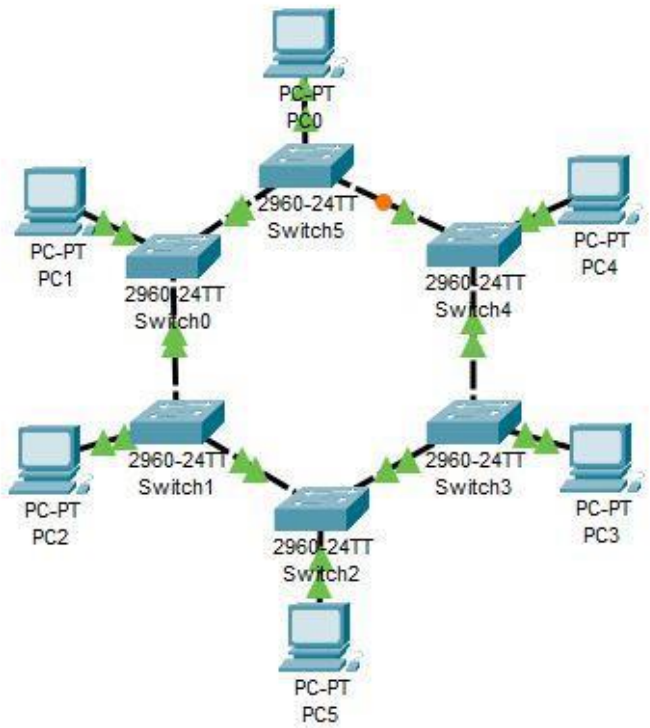
Discussion regarding various topologies.

1. Bus topology
2. Ring topology
3. Star topology
4. Mesh topology
5. Hybrid topology

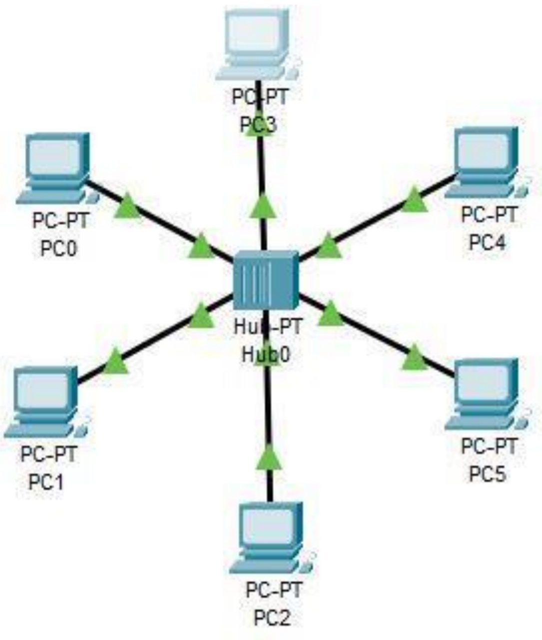
Task 1: Implement the following bus topology using six computers in Cisco packet Tracer (CPT). In the bus topology, devices are connected to a central cable, called bus or backbone. Terminators at each end of the bus stops signal and keeps it from traveling backward.



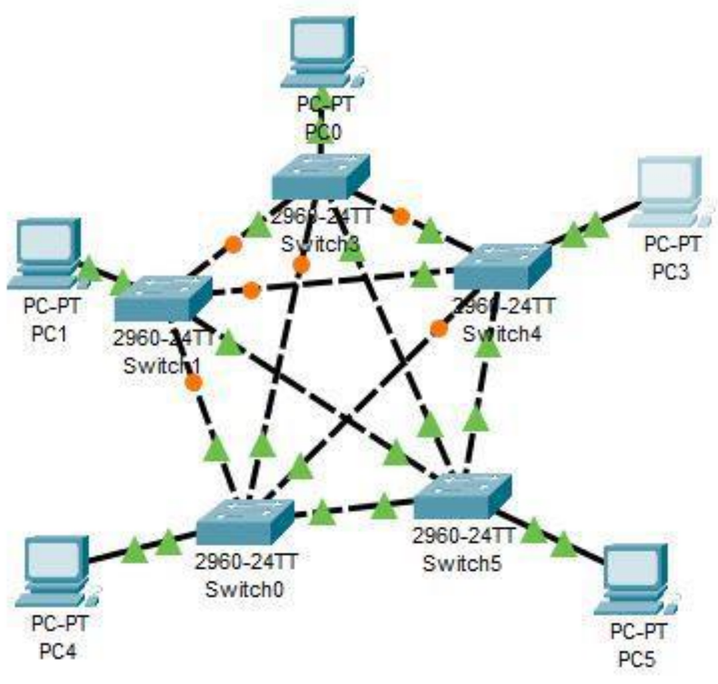
Task 2: Implement the following ring topology using six computers in CPT. In a ring network, every device has exactly two neighboring devices for communication purpose. It is called a ring topology as its formation is like a ring. In this topology, every computer is connected to another computer. Here, the last node is combined with a first one.



Task 3: Implement the following star topology using six computers in CPT. In the star topology, all the computers connect with the help of a hub. This cable is called a central node, and all other nodes are connected using this central node. It is most popular on LAN networks as they are inexpensive and easy to install.



Task 4: Implement the following mesh topology using five computers in CPT. The mesh topology has a unique network design in which each computer on the network connects to every other. It is developing a P2P (point-to-point) connection between all the devices of the network. It offers a high level of redundancy, so even if one network cable fails, still data has an alternative path to reach its destination.



Task 5: Implement the following hybrid topology using six computers in CPT. Hybrid topology is comprising of two or more different types of topologies. It is a reliable and scalable topology, but simultaneously, it is a costly one. It receives the merits and demerits of the topologies used to build it.

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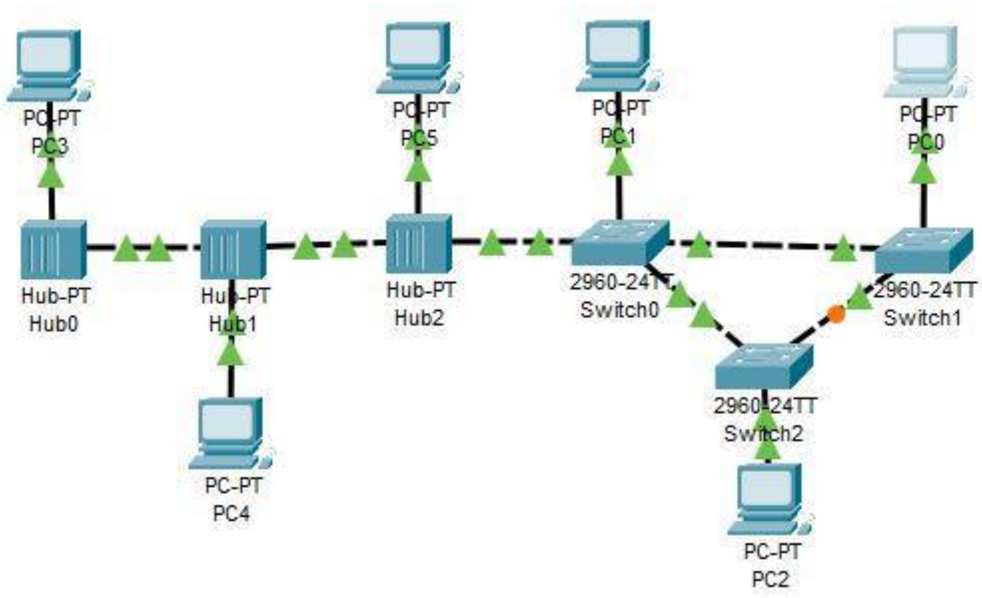

Note: Assign IP address and subnet mask to each computer and run the ping command to check the reachability of the systems. You can use hub or switch as a connecting device to connect the devices. Send messages between source and destination computers and observe the flow of the messages using hub or switch.